

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.*

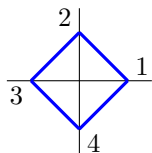
1. Let $a, b, c \in \mathbb{Z}$. Prove the following:
 - a. If $a|b$ and $b|c$ then $a|c$.
 - b. If $a|b$ and $a|c$ then $a|b \pm c$.
2. Show that the set of integers, $\mathbb{Z}$, is a group under addition.
3. Prove that the set of even integers is a group under addition.
4. Show that the following are *not* groups under the given operation.
 - a. $\mathbb{Z}$ under multiplication.
 - b. $S = \{z | z \in \mathbb{Z} \text{ and } z = \sqrt{5}\}$ under addition.
 - c. $\mathbb{Q}$, the set of rationals under multiplication.
 - d. The set of odd integers under addition.
5. Show that $\{1, 2, 3\}$ under multiplication modulo 4 is *not* a group but that $\{1, 2, 3, 4\}$ under multiplication modulo 5 is a group. (Example: $2 \cdot 3 = 6$ and $6 = 2 \pmod{4}$ so $[6] = [2]$ and thus $2 \cdot 3$ is in $\{1, 2, 3\}$).
6. Prove that the set of even integers is a group under addition. (I mistakenly repeated problem 3. For a more challenging problem you might try showing that integers of n multiples for $n \in \mathbb{Z}^+$ are a group under addition)
7. Let $S = \{a + b\sqrt{5} | a, b \text{ are not both zero and } a, b \in \mathbb{Q}\}$. Prove that S is a group. (Hint: rationalize the denominator to show that the inverse of $a + b\sqrt{5}$ is in S .)
8. Prove that the set of all 2×2 matrices with determinant 1 and entries from $\mathbb{R}$ is a group under matrix multiplication.
9. Suppose that if G is a group such that if $a, b, c \in G$ and $ab = ca$ then $b = c$. Prove that G is an Abelian group, i.e., that $ab = ba$ for all $a, b \in G$. Hint: consider the element $aba = aba$.
10. Prove that G is an Abelian group if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.

Presentation problems.

1. Suppose that $a \neq 0$, $b \neq 0$, and $\text{lcm}(a, b) = m$. Prove that if n is a common multiple of a and b then $m|n$.
2. Let $a \neq 0$, $b \neq 0$, and $d = \text{gcd}(a, b)$. Prove that if d' is any common divisor of a and b then $d'|d$.

Theorem. The Fundamental Theorem of Arithmetic. *Every integer greater than 1 is either a prime or the unique product of primes.*

3. If $c^2 = ab$ and $\text{gcd}(a, b) = 1$, prove that a and b are perfect squares (Hint: By the theorem above, a and b can be written as the product of primes. They are perfect squares iff. each prime in this prime factorization has an even power).
4. Prove that if in a group G that $(ab)^2 = a^2b^2$ then G is Abelian.
5. Consider a square placed on Cartesian plane such that the square's center is at the origin and its corners are on the axes (like a diamond; see the figure below). Any motion of the square which keeps the center at the origin and corners on the axes has the same result as one of eight motions: $r_0 \equiv$ rotation of 0° , $r_1 \equiv$ rotation of 90° , $r_3 \equiv$ rotation of 180° , $d_x \equiv x$ -axis reflection, $h_+ \equiv y = x$ reflection, or $h_- \equiv y = -x$ reflection.



Show that $D_4 = \{r_0, r_1, r_3, d_x, d_y, h_+, h_-\}$ is a group under composition of these motions. The set D_4 is called the *dihedral group of degree 4*.

Definition. Let a and n be integers with $n > 0$. The *congruence class of a modulo n* (denoted $[a]$) is the set of integers that are congruent to a modulo n , i.e.,

$$[a] = \{b | b \in \mathbb{Z} \text{ and } b = a \pmod{n}\}$$

For example, the congruence class of 3 modulo 5 is $[3] = \{\dots, -2, 3, 8, 13, \dots\}$.

Definition. The set of all congruence classes modulo n is denoted $\mathbb{Z}_n$ (read “ Z mod n ”).

- $|\mathbb{Z}_n| = n$.
- Addition in $\mathbb{Z}_n$ is defined as $[a] + [b] = [a + b]$.
- Multiplication in $\mathbb{Z}_n$ is defined as $[a] \cdot [b] = [a \cdot b]$.

For example, $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$ for convenience we write $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. $2 + 3 = 5$ and $5 = 1 \pmod{4}$ so $[5] = [1]$.

6. Show that $\mathbb{Z}_2 \times \mathbb{Z}_3$ is a group under modular addition.

Definition. A subset H of a group G is a *subgroup* of G if H is also a group under the operation in G .

7. Let $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ and $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. $\mathbb{Z}_6 \times \mathbb{Z}_4$ is a group under modular addition. Show that $H = \{(0, 0), (3, 0), (0, 2), (3, 2)\}$ is a subgroup.
- ⊙. Two sets are said to be *equal* if there exists a bijective function between them; we write $|A| = |B|$. B is said to have *strictly greater cardinality* than A if there is a one-to-one function from A into B , but there does not exist a bijective function from A into B ; we write $|A| < |B|$. Prove that $|\mathbb{N}| < |\mathbb{R}|$.